

STATISTICS ASSIGNMENT 1 – COMPLETE SOLUTIONS

Measures of Central Tendency & Dispersion • Step-by-Step Solved with Formulae & Calculation Tables

📌 Question 1: Find Mean, Median and Mode

Problem Statement: Find the mean, median and mode from the following data:

Class	0–10	10–20	20–30	30–40	40–50	50–60
Frequency	4	6	20	10	7	3

💡 Easy Guide / आसान तरीका:

1. **Mean** ($\bar{x}$) ke liye mid-value x_i nikalo, $f_i x_i$ ka sum karo aur N se divide karo.
2. **Median** ke liye $\frac{N}{2} = 25$ dhoondo aur median class formula use karo.
3. **Mode** ke liye sabse badi frequency (20) wali modal class chuno aur formula lagao.

1 Step 1: Construct Calculation Table

Class Interval	Frequency (f_i)	Mid-point (x_i)	$f_i x_i$	Cumulative Freq (cf)
0–10	4	5	20	4
10–20	6	15	90	10
20–30	20	25	500	30
30–40	10	35	350	40
40–50	7	45	315	47
50–60	3	55	165	50
Total	$N = 50$	—	$\sum f_i x_i = 1440$	—

2 Step 2: Calculate Arithmetic Mean ($\bar{x}$)

$$\bar{x} = \frac{\sum f_i x_i}{N}$$
$$\bar{x} = \frac{1440}{50} = 28.8$$

3 Step 3: Calculate Median

1. Position of Median: $\frac{N}{2} = \frac{50}{2} = 25$.
2. Cumulative frequency just greater than or equal to 25 is **30** $\Rightarrow$ **Median Class = 20–30**.
 - Lower limit of median class (L) = 20
 - cf of preceding class (cf) = 10
 - Frequency of median class (f) = 20
 - Class length (h) = 10

$$\text{Median} = L + \left(\frac{\frac{N}{2} - cf}{f} \right) \times h$$

$$\text{Median} = 20 + \left(\frac{25 - 10}{20} \right) \times 10 = 20 + \frac{15}{2} = 20 + 7.5 = \mathbf{27.5}$$

4 Step 4: Calculate Mode

Maximum frequency is 20 $\implies$ **Modal Class = 20–30.**

- $L = 20$
- Frequency of modal class (f_1) = 20
- Frequency of previous class (f_0) = 6
- Frequency of next class (f_2) = 10
- Class width (h) = 10

$$\text{Mode} = L + \left(\frac{f_1 - f_0}{2f_1 - f_0 - f_2} \right) \times h$$

$$\text{Mode} = 20 + \left(\frac{20 - 6}{2(20) - 6 - 10} \right) \times 10 = 20 + \left(\frac{14}{40 - 16} \right) \times 10$$

$$\text{Mode} = 20 + \frac{140}{24} = 20 + 5.833 = \mathbf{25.833} \quad \left(\text{or } 25\frac{5}{6} \right)$$

✅ Final Answer:

Mean ($\bar{x}$) = **28.8**

Median = **27.5**

Mode = **25.833** (or $25\frac{5}{6}$)

📌 Question 2: Find D_5 and D_9

Problem Statement: Find D_5 and D_9 for the following data:

Class	0–4	4–8	8–12	12–16	16–20	20–24	24–28	28–32
Frequency	10	12	8	7	5	8	4	6

💡 Easy Guide / आसान तरीका:

Deciles data ko 10 equal parts me divide karte hain.

D_k ka formula: $L + \left(\frac{\frac{kN}{10} - cf}{f} \right) \times h$.

D_5 ke liye position $\frac{5N}{10}$ aur D_9 ke liye $\frac{9N}{10}$ nikalte hain.

1 Step 1: Construct Cumulative Frequency Table

Class Interval	Frequency (f)	Cumulative Frequency (cf)
0–4	10	10
4–8	12	22
8–12	8	30
12–16	7	37
16–20	5	42
20–24	8	50
24–28	4	54
28–32	6	60
Total	$N = 60$	—

2 Step 2: Calculate 5th Decile (D_5)

- Position of D_5 : $\frac{5N}{10} = \frac{5 \times 60}{10} = 30$.
- Looking at the cf column, 30 lies exactly at the end of class **8–12**.
 - Lower limit (L) = 8
 - Cumulative frequency of preceding class (cf) = 22
 - Frequency of class (f) = 8
 - Class width (h) = 4

$$D_5 = L + \left(\frac{\frac{5N}{10} - cf}{f} \right) \times h$$

$$D_5 = 8 + \left(\frac{30 - 22}{8} \right) \times 4 = 8 + \left(\frac{8}{8} \right) \times 4 = 8 + 4 = 12$$

3 Step 3: Calculate 9th Decile (D_9)

- Position of D_9 : $\frac{9N}{10} = \frac{9 \times 60}{10} = 54$.

2. In the cf column, 54 lies in the class **24–28**.

- Lower limit (L) = 24
- cf of previous class (cf) = 50
- Frequency of class (f) = 4
- Class width (h) = 4

$$D_9 = L + \left(\frac{\frac{9N}{10} - cf}{f} \right) \times h$$

$$D_9 = 24 + \left(\frac{54 - 50}{4} \right) \times 4 = 24 + \left(\frac{4}{4} \right) \times 4 = 24 + 4 = \mathbf{28}$$

✔ **Final Answer:**

5th Decile (D_5) = **12**

9th Decile (D_9) = **28**

📌 Question 3: Mean Deviation from Mean & Co-efficient

Problem Statement: Find the mean deviation from mean and co-efficient of mean deviation for the following data:

Class	2-4	4-6	6-8	8-10
Frequency	3	4	2	1

💡 Easy Guide / आसान तरीका:

1. Pehle Mean $\bar{x} = \frac{\sum f_i x_i}{N}$ nikalo.
2. Har class ke mid-point x_i ka Mean se difference nikalo: $|x_i - \bar{x}|$ (signs ignore karna hai).
3. Us difference ko frequency f_i se multiply karke sum karo aur N se divide karo: $MD(\bar{x}) = \frac{\sum f_i |x_i - \bar{x}|}{N}$.
4. Coefficient = $\frac{MD(\bar{x})}{\bar{x}}$.

1 Step 1: Compute Mean ($\bar{x}$) and Absolute Deviations Table

Class	Frequency (f_i)	Mid-point (x_i)	$f_i x_i$	$ x_i - \bar{x} = x_i - 5.2 $	$f_i x_i - \bar{x} $
2-4	3	3	9	$ 3 - 5.2 = 2.2$	$3 \times 2.2 = 6.6$
4-6	4	5	20	$ 5 - 5.2 = 0.2$	$4 \times 0.2 = 0.8$
6-8	2	7	14	$ 7 - 5.2 = 1.8$	$2 \times 1.8 = 3.6$
8-10	1	9	9	$ 9 - 5.2 = 3.8$	$1 \times 3.8 = 3.8$
Total	$N = 10$	—	$\sum f_i x_i = 52$	—	$\sum f_i x_i - \bar{x} = 14.8$

2 Step 2: Mean Calculation

$$\bar{x} = \frac{\sum f_i x_i}{N} = \frac{52}{10} = 5.2$$

3 Step 3: Calculate Mean Deviation from Mean

$$MD(\bar{x}) = \frac{\sum f_i |x_i - \bar{x}|}{N} = \frac{14.8}{10} = 1.48$$

4 Step 4: Calculate Coefficient of Mean Deviation

$$\text{Coefficient of } MD(\bar{x}) = \frac{MD(\bar{x})}{\bar{x}} = \frac{1.48}{5.2} = \frac{37}{130} \approx 0.2846 \quad (\text{or } 28.46\%)$$

✅ Final Answer:

Mean ($\bar{x}$) = 5.2

Mean Deviation from Mean = **1.48**

Coefficient of Mean Deviation = **0.2846** (or **28.46%**)

Question 4: Calculate Coefficient of Variation (C.V.)

Problem Statement: Calculate the co-efficient of variation (C.V.) from the following data:

Age (in years)	0-9	10-19	20-29	30-39	40-49	50-59
No. of persons (f)	13	38	67	76	22	4

💡 Easy Guide / आसान तरीका:

- Inclusive class intervals (0-9, 10-19) ke mid-values hote hain: $\frac{0+9}{2} = 4.5, 14.5, 24.5 \dots$
- Step-deviation method se calculation easy ho jaati hai: $A = 24.5, h = 10, u_i = \frac{x_i - 24.5}{10}$.
- Formula: $C.V. = \frac{\sigma}{\bar{x}} \times 100$.

1 Step 1: Step-Deviation Calculation Table

Let Assumed Mean $A = 24.5$, width $h = 10$. $u_i = \frac{x_i - 24.5}{10}$.

Age (Class)	f_i	Mid-point (x_i)	u_i	$f_i u_i$	u_i^2	$f_i u_i^2$
0-9	13	4.5	-2	-26	4	52
10-19	38	14.5	-1	-38	1	38
20-29	67	24.5	0	0	0	0
30-39	76	34.5	+1	+76	1	76
40-49	22	44.5	+2	+44	4	88
50-59	4	54.5	+3	+12	9	36
Total	$N = 220$	—	—	$\sum f_i u_i = 68$	—	$\sum f_i u_i^2 = 290$

2 Step 2: Calculate Mean ($\bar{x}$)

$$\bar{x} = A + \left(\frac{\sum f_i u_i}{N} \right) \times h$$
$$\bar{x} = 24.5 + \left(\frac{68}{220} \right) \times 10 = 24.5 + 3.0909 = \mathbf{27.5909 \text{ years}}$$

3 Step 3: Calculate Standard Deviation (σ)

$$\sigma = h \times \sqrt{\frac{\sum f_i u_i^2}{N} - \left(\frac{\sum f_i u_i}{N} \right)^2}$$
$$\frac{\sum f_i u_i^2}{N} = \frac{290}{220} \approx 1.318182$$
$$\left(\frac{\sum f_i u_i}{N} \right)^2 = \left(\frac{68}{220} \right)^2 = (0.309091)^2 \approx 0.095537$$
$$\text{Variance in } u = 1.318182 - 0.095537 = 1.222645$$

$$\sigma = 10 \times \sqrt{1.222645} = 10 \times 1.105733 = \mathbf{11.0573 \text{ years}}$$

4 Step 4: Calculate Coefficient of Variation (C.V.)

$$C.V. = \frac{\sigma}{\bar{x}} \times 100$$

$$C.V. = \frac{11.0573}{27.5909} \times 100 \approx \mathbf{40.08\%}$$

✅ Final Answer:

Mean ($\bar{x}$) = **27.59 years**

Standard Deviation (σ) = **11.06 years**

Coefficient of Variation (C.V.) = **40.08%**

📌 Question 5: Find Q_1 and Q_3

Problem Statement: Find Q_1 and Q_3 from the following data:

Class	20–25	25–30	30–35	35–40	40–45	45–50
Frequency (f)	5	12	15	8	5	5

💡 Easy Guide / आसान तरीका:

- Q_1 (Lower Quartile) = data ka 25% mark $\implies$ position $\frac{N}{4}$.
- Q_3 (Upper Quartile) = data ka 75% mark $\implies$ position $\frac{3N}{4}$.
- Formula: $Q_k = L + \left(\frac{\frac{k \cdot N}{4} - cf}{f} \right) \times h$.

1 Step 1: Construct Cumulative Frequency Table

Class Interval	Frequency (f)	Cumulative Frequency (cf)
20–25	5	5
25–30	12	17
30–35	15	32
35–40	8	40
40–45	5	45
45–50	5	50
Total	$N = 50$	—

2 Step 2: Calculate First Quartile (Q_1)

- Position of Q_1 : $\frac{N}{4} = \frac{50}{4} = 12.5$.
- cf just greater than 12.5 is **17** $\implies Q_1$ **Class = 25–30**.

- Lower limit (L) = 25
- Preceding cf (cf) = 5
- Frequency of class (f) = 12
- Class length (h) = 5

$$Q_1 = L + \left(\frac{\frac{N}{4} - cf}{f} \right) \times h$$

$$Q_1 = 25 + \left(\frac{12.5 - 5}{12} \right) \times 5 = 25 + \left(\frac{7.5}{12} \right) \times 5 = 25 + \frac{37.5}{12} = 25 + 3.125 = \mathbf{28.125}$$

3 Step 3: Calculate Third Quartile (Q_3)

- Position of Q_3 : $\frac{3N}{4} = \frac{3 \times 50}{4} = 37.5$.
- cf just greater than 37.5 is **40** $\implies Q_3$ **Class = 35–40**.

- Lower limit (L) = 35

- Preceding cf (cf) = 32
- Frequency of class (f) = 8
- Class length (h) = 5

$$Q_3 = L + \left(\frac{\frac{3N}{4} - cf}{f} \right) \times h$$

$$Q_3 = 35 + \left(\frac{37.5 - 32}{8} \right) \times 5 = 35 + \left(\frac{5.5}{8} \right) \times 5 = 35 + \frac{27.5}{8} = 35 + 3.4375 = \mathbf{38.4375}$$

 Final Answer:

First Quartile (Q_1) = **28.125**

Third Quartile (Q_3) = **38.4375**

Question 6: Find Q_1 , D_8 , and P_{65}

Problem Statement: Find Q_1 , D_8 , and P_{65} from the following data:

Class	20–30	30–40	40–50	50–60	60–70	70–80	80–90
Frequency	69	167	207	65	58	27	10

💡 Easy Guide / आसान तरीका:

- $Q_1 = \text{Position } \frac{N}{4} = \frac{603}{4} = 150.75$.

- $D_8 = \text{Position } \frac{8N}{10} = \frac{4824}{10} = 482.4$.

- $P_{65} = \text{Position } \frac{65N}{100} = 391.95$.

Har partition value ke liye corresponding class dhoond kar standard interpolation formula lagana hai.

1 Step 1: Cumulative Frequency Table

Class Interval	Frequency (f)	Cumulative Frequency (cf)
20–30	69	69
30–40	167	236
40–50	207	443
50–60	65	508
60–70	58	566
70–80	27	593
80–90	10	603
Total	$N = 603$	—

2 Step 2: Calculate First Quartile (Q_1)

Position: $\frac{N}{4} = 150.75 \implies$ Class **30–40** ($L = 30, cf = 69, f = 167, h = 10$).

$$Q_1 = 30 + \left(\frac{150.75 - 69}{167} \right) \times 10 = 30 + \left(\frac{81.75}{167} \right) \times 10 = 30 + 4.8952 = \mathbf{34.8952}$$

3 Step 3: Calculate 8th Decile (D_8)

Position: $\frac{8N}{10} = 482.4 \implies$ Class **50–60** ($L = 50, cf = 443, f = 65, h = 10$).

$$D_8 = 50 + \left(\frac{482.4 - 443}{65} \right) \times 10 = 50 + \left(\frac{39.4}{65} \right) \times 10 = 50 + 6.0615 = \mathbf{56.0615}$$

4 Step 4: Calculate 65th Percentile (P_{65})

Position: $\frac{65N}{100} = 391.95 \implies$ Class **40–50** ($L = 40, cf = 236, f = 207, h = 10$).

$$P_{65} = 40 + \left(\frac{391.95 - 236}{207} \right) \times 10 = 40 + \left(\frac{155.95}{207} \right) \times 10 = 40 + 7.5338 = \mathbf{47.5338}$$

✅ **Final Answer:**

First Quartile (Q_1) = **34.90**

Eighth Decile (D_8) = **56.06**

65th Percentile (P_{65}) = **47.53**

📌 Question 7: Mean Deviation about Mean and Median

Problem Statement: Find the mean deviation about mean and mean deviation about median for the following data:
Data values: 3, 9, 5, 3, 12, 10, 18, 4, 7, 19, 21

💡 Easy Guide / आसान तरीका:

- Total items $n = 11$.
- **Part 1:** Sum nikalo $\sum x = 111$, Mean $= \frac{111}{11} = 10.0909$. Fir har value ka $|x - \bar{x}|$ nikaal kar average lo.
- **Part 2:** Values ko ascending order me arrange karo. Middle value (6th term) Median hogi $= 9$. Fir $|x - 9|$ nikaal kar average lo.

1 Part A: Mean Deviation about Mean

$$\sum x_i = 3 + 9 + 5 + 3 + 12 + 10 + 18 + 4 + 7 + 19 + 21 = 111$$

$$\bar{x} = \frac{111}{11} \approx 10.0909$$

Absolute deviations from Mean $|x_i - 10.0909|$:

x_i	$ x_i - \bar{x} $	x_i	$ x_i - \bar{x} $
3	7.0909	18	7.9091
9	1.0909	4	6.0909
5	5.0909	7	3.0909
3	7.0909	19	8.9091
12	1.9091	21	10.9091
10	0.0909	Total	$\sum x_i - \bar{x} = 59.2727$

$$MD(\bar{x}) = \frac{\sum |x_i - \bar{x}|}{n} = \frac{59.2727}{11} \approx \mathbf{5.3884}$$

2 Part B: Mean Deviation about Median

Arranging data in ascending order:

$$3, 3, 4, 5, 7, \mathbf{9}, 10, 12, 18, 19, 21$$

Since $n = 11$ (odd):

$$\text{Median } (M) = \left(\frac{11+1}{2} \right)\text{-th term} = 6\text{-th term} = \mathbf{9}$$

Absolute deviations from Median $|x_i - 9|$:

x_i	$ x_i - 9 $	x_i	$ x_i - 9 $
3	$ 3 - 9 = 6$	10	$ 10 - 9 = 1$
3	$ 3 - 9 = 6$	12	$ 12 - 9 = 3$
4	$ 4 - 9 = 5$	18	$ 18 - 9 = 9$
5	$ 5 - 9 = 4$	19	$ 19 - 9 = 10$

x_i	$ x_i - 9 $	x_i	$ x_i - 9 $
7	$ 7 - 9 = 2$	21	$ 21 - 9 = 12$
9	$ 9 - 9 = 0$	Total	$\sum x_i - M = 58$

$$MD(M) = \frac{\sum |x_i - M|}{n} = \frac{58}{11} \approx \mathbf{5.2727}$$

 Final Answer:

Mean = **10.09**

Mean Deviation about Mean = **5.39**

Median = **9**

Mean Deviation about Median = **5.27** (or $\frac{58}{11}$)

📌 Question 8: Mean Deviation about Median

Problem Statement: Find the mean deviation about median for the following data:

x_i	3	6	9	12	13	15	21	22
f_i	3	4	5	2	4	5	4	3

💡 Easy Guide / आसान तरीका:

1. Pehle cumulative frequency cf banakar Median M find karte hain ($N/2 = 15$).
2. $cf \geq 15$ corresponds to $x = 13$, so Median = 13.
3. Fir har x_i ke liye $|x_i - 13|$ nikalo, usko f_i se multiply karo aur sum ko N se divide karo.

1 Step 1: Determine Median and Cumulative Frequency

Total Frequency $N = \sum f_i = 30$.

Median position = $\frac{N}{2} = 15$. In the cf column, $cf \geq 15$ is **18**, which corresponds to $x_i = 13$.

$$\text{Median } (M) = 13$$

2 Step 2: Table of Absolute Deviations

x_i	f_i	cf	$ x_i - M = x_i - 13 $	$f_i x_i - M $
3	3	3	$ 3 - 13 = 10$	$3 \times 10 = 30$
6	4	7	$ 6 - 13 = 7$	$4 \times 7 = 28$
9	5	12	$ 9 - 13 = 4$	$5 \times 4 = 20$
12	2	14	$ 12 - 13 = 1$	$2 \times 1 = 2$
13	4	18	$ 13 - 13 = 0$	$4 \times 0 = 0$
15	5	23	$ 15 - 13 = 2$	$5 \times 2 = 10$
21	4	27	$ 21 - 13 = 8$	$4 \times 8 = 32$
22	3	30	$ 22 - 13 = 9$	$3 \times 9 = 27$
Total	$N = 30$	—	—	$\sum f_i x_i - M = 149$

3 Step 3: Calculate Mean Deviation about Median

$$MD(\text{Median}) = \frac{\sum f_i|x_i - M|}{N} = \frac{149}{30} \approx 4.9667$$

✅ Final Answer:

Median (M) = **13**

Mean Deviation about Median = **4.97** (or $\frac{149}{30}$)

📌 Question 9: Find D_6 , P_{20} and P_{100}

Problem Statement: Find D_6 , P_{20} and P_{100} from the following data:

Class	0–10	10–20	20–30	30–40	40–50	50–60
Frequency	5	8	21	8	5	4

💡 Easy Guide / आसान तरीका:

- Total frequency $N = 51$.
- D_6 position = $\frac{6 \times 51}{10} = 30.6$.
- P_{20} position = $\frac{20 \times 51}{100} = 10.2$.
- P_{100} position = 51 (jo distribution ki upper limit 60 ko represent karta hai).

1 Step 1: Cumulative Frequency Table

Class Interval	Frequency (f)	Cumulative Frequency (cf)
0–10	5	5
10–20	8	13
20–30	21	34
30–40	8	42
40–50	5	47
50–60	4	51
Total	$N = 51$	—

2 Step 2: Calculate 6th Decile (D_6)

Position: $\frac{6N}{10} = 30.6 \implies$ Class **20–30** ($L = 20, cf = 13, f = 21, h = 10$).

$$D_6 = 20 + \left(\frac{30.6 - 13}{21} \right) \times 10 = 20 + \left(\frac{17.6}{21} \right) \times 10 = 20 + \frac{176}{21} \approx \mathbf{28.3810}$$

3 Step 3: Calculate 20th Percentile (P_{20})

Position: $\frac{20N}{100} = 10.2 \implies$ Class **10–20** ($L = 10, cf = 5, f = 8, h = 10$).

$$P_{20} = 10 + \left(\frac{10.2 - 5}{8} \right) \times 10 = 10 + \left(\frac{5.2}{8} \right) \times 10 = 10 + 6.5 = \mathbf{16.5}$$

4 Step 4: Calculate 100th Percentile (P_{100})

Position: $\frac{100N}{100} = 51 \implies$ Highest Class **50–60** ($L = 50, cf = 47, f = 4, h = 10$).

$$P_{100} = 50 + \left(\frac{51 - 47}{4} \right) \times 10 = 50 + \left(\frac{4}{4} \right) \times 10 = 50 + 10 = \mathbf{60}$$

(Note: 100th percentile is the maximum boundary value of the distribution = 60)

✅ **Final Answer:**

$$D_6 = \mathbf{28.38}$$

$$P_{20} = \mathbf{16.5}$$

$$P_{100} = \mathbf{60}$$

Question 10: Find Missing Frequencies f_1 and f_2

Problem Statement: Find the missing frequencies f_1 and f_2 of the following distribution, given that the **mean is 21.4** and **total frequency is 40**:

Class-Interval	0-8	8-16	16-24	24-32	32-40
Frequency	6	f_1	10	f_2	9

Easy Guide / आसान तरीका:

Yahan 2 unknowns (f_1 aur f_2) hain, isliye 2 equations banengi:

- **Equation 1:** Total frequency = 40 $\implies f_1 + f_2 = 15$.

- **Equation 2:** Mean = 21.4 $\implies \frac{\sum f_i x_i}{40} = 21.4$.

Dono ko solve karke exact integer answers mil jaate hain!

1 Step 1: Construct Calculation Table with Mid-values

Class Interval	Frequency (f_i)	Mid-point (x_i)	$f_i x_i$
0-8	6	4	$6 \times 4 = 24$
8-16	f_1	12	$12f_1$
16-24	10	20	$10 \times 20 = 200$
24-32	f_2	28	$28f_2$
32-40	9	36	$9 \times 36 = 324$
Total	$N = 40$	—	$\sum f_i x_i = 548 + 12f_1 + 28f_2$

2 Step 2: Form Equation (1) from Total Frequency

$$6 + f_1 + 10 + f_2 + 9 = 40$$

$$25 + f_1 + f_2 = 40$$

$$f_1 + f_2 = 15 \quad \text{--- [Equation 1]}$$

3 Step 3: Form Equation (2) from Given Mean (21.4)

$$\bar{x} = \frac{\sum f_i x_i}{N} = 21.4$$

$$\frac{548 + 12f_1 + 28f_2}{40} = 21.4$$

$$548 + 12f_1 + 28f_2 = 40 \times 21.4 = 856$$

$$12f_1 + 28f_2 = 856 - 548 = 308$$

Divide entire equation by 4:

$$3f_1 + 7f_2 = 77 \quad \text{--- [Equation 2]}$$

4 Step 4: Solve the Equations Simultaneously

From Equation 1: $f_1 = 15 - f_2$. Substitute into Equation 2:

$$3(15 - f_2) + 7f_2 = 77$$

$$45 - 3f_2 + 7f_2 = 77$$

$$4f_2 = 77 - 45 = 32$$

$$f_2 = \frac{32}{4} = 8$$

Now calculate f_1 :

$$f_1 = 15 - f_2 = 15 - 8 = 7$$

✅ Final Answer:

$$f_1 = 7$$

$$f_2 = 8$$

📌 Question 11: Find Missing Frequencies f_1 and f_2

Problem Statement: Find the missing frequencies f_1 and f_2 for the following data whose **mean is 1.46** and **total frequency is 200**:

x_i	0	1	2	3	4	5
f_i	46	f_1	f_2	25	10	5

💡 Easy Guide / आसान तरीका:

- Yeh discrete series hai (x_i direct values hain).
- Total frequency $N = 200 \implies f_1 + f_2 = 114$.
- Mean $\bar{x} = 1.46 \implies \sum f_i x_i = 200 \times 1.46 = 292$.
- Dono linear equations ko subtract karke f_1 aur f_2 aasani se mil jate hain.

1 Step 1: Construct Calculation Table

x_i	f_i	$f_i x_i$
0	46	$46 \times 0 = 0$
1	f_1	$f_1 \times 1 = f_1$
2	f_2	$f_2 \times 2 = 2f_2$
3	25	$25 \times 3 = 75$
4	10	$10 \times 4 = 40$
5	5	$5 \times 5 = 25$
Total	$N = 200$	$\sum f_i x_i = 140 + f_1 + 2f_2$

2 Step 2: Form Equation (1) from Total Frequency

$$46 + f_1 + f_2 + 25 + 10 + 5 = 200$$

$$86 + f_1 + f_2 = 200$$

$$f_1 + f_2 = 114 \quad \text{--- [Equation 1]}$$

3 Step 3: Form Equation (2) from Mean (1.46)

$$\bar{x} = \frac{\sum f_i x_i}{N} = 1.46$$

$$\frac{140 + f_1 + 2f_2}{200} = 1.46$$

$$140 + f_1 + 2f_2 = 200 \times 1.46 = 292$$

$$f_1 + 2f_2 = 292 - 140 = 152 \quad \text{--- [Equation 2]}$$

4**Step 4: Solve Simultaneous Equations**

Subtract Equation 1 from Equation 2:

$$(f_1 + 2f_2) - (f_1 + f_2) = 152 - 114$$
$$f_2 = \mathbf{38}$$

Substitute $f_2 = 38$ into Equation 1:

$$f_1 + 38 = 114 \implies f_1 = 114 - 38 = \mathbf{76}$$

 **Final Answer:**

$$f_1 = \mathbf{76}$$

$$f_2 = \mathbf{38}$$

📌 Question 12: Find the Harmonic Mean

Problem Statement: Find the Harmonic Mean for the following distribution:

Marks (x_i)	20	30	40	50	60	70	80
Frequency (f_i)	3	61	132	153	140	51	3

💡 **Easy Guide / आसान तरीका:**

Harmonic Mean ka formula: $HM = \frac{N}{\sum \left(\frac{f_i}{x_i} \right)}$.

Har frequency ko uske marks se divide karo, unka sum nikalo, fir total frequency $N = 543$ ko us sum se divide kar do.

1 Step 1: Compute $\frac{f_i}{x_i}$ Table

Marks (x_i)	Frequency (f_i)	$\frac{f_i}{x_i}$ (Fraction)	$\frac{f_i}{x_i}$ (Decimal)
20	3	$\frac{3}{20}$	0.150000
30	61	$\frac{61}{30}$	2.033333
40	132	$\frac{132}{40}$	3.300000
50	153	$\frac{153}{50}$	3.060000
60	140	$\frac{140}{60}$	2.333333
70	51	$\frac{51}{70}$	0.728571
80	3	$\frac{3}{80}$	0.037500
Total	$N = 543$	—	$\sum \frac{f_i}{x_i} \approx 11.642738$

2 Step 2: Harmonic Mean Formula & Calculation

$$HM = \frac{N}{\sum \left(\frac{f_i}{x_i} \right)}$$

$$HM = \frac{543}{11.642738} \approx \mathbf{46.6385}$$

Rounding to two decimal places:

$$HM \approx \mathbf{46.64}$$

✅ **Final Answer:**

Total Frequency (N) = **543**

$\sum (f_i/x_i) = \mathbf{11.6427}$

Harmonic Mean (H.M.) = **46.64**

Question 13: Statistical Theory & Definitions

Problem Statement:

- Define Range.
- Define Mean Deviation.
- Define Standard Deviation.
- State Measures of Dispersion.
- State Merits and Demerits of Mean, Median, and Mode.

1. Definition of Range

Definition: Range is the simplest and most elementary measure of dispersion. It is defined as the absolute difference between the largest (maximum) observation and the smallest (minimum) observation present in a given dataset.

$$\text{Range} = L - S$$

$$\text{Coefficient of Range} = \frac{L - S}{L + S}$$

where L = Largest value, S = Smallest value.

2. Definition of Mean Deviation

Definition: Mean Deviation (or Average Deviation) is the arithmetic mean of the absolute values of the numerical deviations of observations taken from an average (usually the arithmetic mean or median). Algebraic signs (+ or -) are deliberately ignored.

$$\text{For Individual Series: } MD = \frac{\sum |x_i - A|}{n}$$

$$\text{For Grouped Series: } MD = \frac{\sum f_i |x_i - A|}{N}$$

$$\text{Coefficient of } MD = \frac{MD}{A}$$

where A is the Mean ($\bar{x}$) or Median (M).

3. Definition of Standard Deviation

Definition: Standard Deviation was introduced by Karl Pearson in 1893. It is defined as the **positive square root of the arithmetic mean of the squared deviations of observations taken from their arithmetic mean**. It is universally denoted by the Greek letter σ (sigma).

$$\sigma = \sqrt{\frac{\sum f_i(x_i - \bar{x})^2}{N}} = \sqrt{\frac{\sum f_i x_i^2}{N} - \left(\frac{\sum f_i x_i}{N}\right)^2}$$

Variance = σ^2

4. State Measures of Dispersion

Dispersion represents the extent to which values in a distribution scatter around a central tendency. They are categorized into:

A. Absolute Measures (Have units)

1. Range ($L - S$)
2. Quartile Deviation ($QD = \frac{Q_3 - Q_1}{2}$)
3. Mean Deviation
4. Standard Deviation & Variance

B. Relative Measures (Unit-free)

1. Coefficient of Range = $\frac{L-S}{L+S}$
2. Coefficient of QD = $\frac{Q_3 - Q_1}{Q_3 + Q_1}$
3. Coefficient of MD = $\frac{MD}{\text{Mean/Median}}$
4. Coefficient of Variation (C.V.) = $\frac{\sigma}{\bar{x}} \times 100$

5. Merits and Demerits of Mean, Median, and Mode

A. Arithmetic Mean ($\bar{x}$)

✓ Merits

- Rigidly defined by a mathematical formula.
- Based on all observations of the dataset.
- Capable of further algebraic treatment (e.g., combined mean).
- Least affected by sampling fluctuations (most stable).

✗ Demerits

- Highly sensitive to extreme values (outliers).
- Cannot be calculated for open-ended classes without assumptions.
- Cannot be determined graphically.
- Cannot be measured for qualitative data (honesty, beauty).

B. Median (M)

✓ Merits

- Easy to understand and compute.
- Completely unaffected by extreme values / outliers.
- Can be calculated for open-ended frequency distributions.
- Can be determined graphically using Ogive curves.
- Ideal measure for qualitative data.

✗ Demerits

- Not based on all items (depends only on middle position).
- Not amenable to algebraic treatment (cannot combine medians).
- Requires sorting/ranking of data (laborious for large data).
- Less stable under sampling fluctuations than mean.

C. Mode (Z)

✓ Merits

- Easiest to understand; can often be determined by inspection.
- Unaffected by extreme values.
- Can be computed for open-ended classes.
- Can be determined graphically using a Histogram.
- Very practical for trade and industry (popular sizes).

✗ Demerits

- Ill-defined in bimodal, multimodal, or uniform distributions.
- Not based on all observations in the dataset.
- Not capable of further algebraic manipulation.
- Most unstable average under sampling fluctuations.

✓ Final Summary:

All theoretical questions answered comprehensively with clear mathematical definitions, formulae, classifications, and side-by-side merit/demerit comparisons.